ACM 104 Final

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1. True or False

a) True

This is guaranteed by Graham-Schmidt process because for any finite dimensional inner product with a dimension $n \geq 1$ we can start with an arbitrary basis and sequentially apply Graham-Schmidt.

b) False

By definition for an orthonormal matrix Q satisfies $Q^T Q = I$

$$\det(Q^T Q) = \det(I)$$
$$\det(Q^T)\det(Q) = 1$$
$$(\det(Q))^2 = 1$$
$$\therefore \det(Q) = \pm 1 \qquad \text{(not only 1)}$$

c) True

Let $Av = \lambda_A v$ and $Bv = \lambda_B v$

$(A+B)v = Av + Bv = \lambda_A v + \lambda_B v = (\lambda_A + \lambda_B) v$

So v is eigenvector of $A+B$ w/ eigenvalue $\lambda_A + \lambda_B$.

d) False by Counterexample:

$A = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$; $\lambda = 1$ is an eigenvalue for both

but not for $A+B = \begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix}$ which has $\lambda_1^* = 2$, $\lambda_2^* = 4$.

e) True

If A is diagonalizable it can be written as $A = PDP^{-1}$ where D is diagonal. $A^n = (PDP^{-1})^n = PD^nP^{-1}$

Since D is diagonal D^n is same just w/ entries to the power n.

So A^n is similar to diagonal matrix D^n and therefore is diagonalizable.

f) False

For a real symmetric matrix eigenvectors for different eigenvalues must be orthogonal. I think this is the Spectral Theorem

$$V_1 \cdot V_2 = 1+2 = 3 \neq 0 \text{ so not orthogonal } \therefore \text{ false}$$

g) False

$$A^TA = \begin{bmatrix} 0 & 0 \\ 0 & 2025^2 \end{bmatrix} \rightarrow \text{singular values } 0 \text{ and } 2025$$

Singular values are $\sqrt{\text{eigenvalues of } A^TA}$

h) False by Counterexample

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix} \text{ eigenvales: } \lambda_1 = 1, \lambda_2 = 2$$

since Singular values are $\sqrt{\text{eigenvalues of } A^TA}$

$$A^TA = \begin{bmatrix} 1 & 1 \\ 1 & 5 \end{bmatrix} \quad \det(A^TA - \mu I) = 0 \rightarrow \mu^2 - 6\mu + 4 = 0$$

$\mu = 1$ isn't a solution

Since 1 isn't an eigenvalue of A^TA, 1 is not a singular value of A.

i) False by Counterexample

$$A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, A^TA = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \rightarrow \text{Singular values: } 1 \text{ and } 0.$$

$$A^2 = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \rightarrow \text{singular values } 0 \text{ and } 0.$$

They are not the same $(1^2 = 0)$ so false.

j) True. Start with SVD of A: $A = U\Sigma V^T$; U, V are orthogonal Σ is diagonal w/ singular values σ_i. $A^+ = V\Sigma^+U^T$; Σ contains entries $1/\sigma_i$ for all non-zero σ_i. ↳ already in SVD form

$$(A^+)^+ = (U^T)^T(\Sigma^+)^+V^T = U\Sigma V^T = A$$

↓

Flip $1/\sigma_i$ $\left(\frac{1}{1/\sigma_i}\right) = \sigma_i$

2. a) Find vector in plane: let $y=0$, $x=1$

$n = \begin{bmatrix} 1 \\ -4 \\ 1 \end{bmatrix}$ $\quad 1-4(0)+z=0 \rightarrow z=-1$, $\quad v_1 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$

$$v_2 = n \times v_1 = \begin{bmatrix} 1 \\ -4 \\ 1 \end{bmatrix} \times \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix};$$

$2-4(1)+2=0 \rightarrow v_2$ in plane ✓ ; $2(1)+1(0)+2(-1)=0 \rightarrow$ orthogonal to v_1 ✓

$\|v_1\| = \sqrt{2}$

$\|v_2\| = 3$

$$\boxed{q_1 = \begin{bmatrix} 1/\sqrt{2} \\ 0 \\ -1/\sqrt{2} \end{bmatrix}, \quad q_2 = \begin{bmatrix} 2/3 \\ 1/3 \\ 2/3 \end{bmatrix}}$$

b) $\text{proj}_P v = (v \cdot q_1) q_1 + (v \cdot q_2) q_2$

$$v \cdot q_1 = \begin{bmatrix} 1 \\ 0 \\ 4 \end{bmatrix} \cdot \begin{bmatrix} 1/\sqrt{2} \\ 0 \\ -1/\sqrt{2} \end{bmatrix} = -\frac{3}{\sqrt{2}} \; ; \quad v \cdot q_2 = \begin{bmatrix} 1 \\ 0 \\ 4 \end{bmatrix} \cdot \begin{bmatrix} 2/3 \\ 1/3 \\ 2/3 \end{bmatrix} = \frac{10}{3}$$

$$\left(-\frac{3}{\sqrt{2}}\right) q_1 = \overrightarrow{\begin{bmatrix} -1.5 \\ 0 \\ 1.5 \end{bmatrix}}^{w_1} ; \quad \left(\frac{10}{3}\right) q_2 = \overrightarrow{\begin{bmatrix} 20/9 \\ 10/9 \\ 20/9 \end{bmatrix}}^{w_2} ; \quad \text{proj}_P v = w_1 + w_2 = \boxed{\begin{bmatrix} 13/18 \\ 10/9 \\ 67/18 \end{bmatrix}}$$

$$x - 4y + z = \frac{13}{18} - 4\,\frac{10}{9} + \frac{67}{18} = 0 \checkmark$$

3. A is like $2J$ just with 1 subtracted from each diagonal entry.

$$A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}; \quad 2J = \begin{bmatrix} 2 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{bmatrix} \rightarrow A = 2J - I$$

$$\vec{1} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \quad ; \quad J\vec{1} = \begin{bmatrix} 1+1+1 \\ 1+1+1 \\ 1+1+1 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \\ 3 \end{bmatrix} = 3\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \therefore J\vec{1} = 3\vec{1}$$

so 3 is an eigenvalue of J w/ eigenvector $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$.

a) Along $\vec{1}$, $J\vec{1} = 3\vec{1}$, so

$$A\vec{1} = (2J - I)\vec{1} = (2 \cdot 3 - 1)\vec{1} = 5\vec{1} \rightarrow \lambda = 5$$

If $x \perp 1$ (i.e. $x_1 + x_2 + x_3 = 0$), $Jx = 0$ so

$$Ax = (2J - I)x = -x$$

So $\lambda = -1$ is an eigenvalue w/ multiplicity 2. $\boxed{\text{Eigenvalues: } 5, -1, -1}$

b) An eigenvector for $\lambda = 5$ is $v_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

We can pick two independent vectors with the sum of entries 0 so they lie in the $\lambda = -1$ eigenspace, say $v_2 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$

Then $Av_2 = -v_2$, $Av_3 = -v_3$

$$\boxed{\text{Eigenbasis: } \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}}$$

c) For $\lambda = 5$, we normalize: $u_1 = \frac{1}{\sqrt{3}}\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

For $\lambda = -1$ we need to orthogonalize v_2, v_3

Let $w_1 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$, $u_2 = \frac{1}{\sqrt{2}}\begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$

$$w_2 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} \times \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix}, \quad u_3 = \frac{1}{\sqrt{6}}\begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix}$$

$$P_1 = u_1 u_1^T = \frac{1}{3}\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

$$P_2 = I - P_1 = \frac{1}{3}\begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix}$$

(eigenspaces span all $\mathbb{R}^3 \rightarrow P_1 + P_2 = I$)

$$\therefore \boxed{A = 5\left[\frac{1}{3}\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}\right] + (-1)\left[\frac{1}{3}\begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix}\right]}$$

4. We can use Gershgorin circle theorem here.

For each row i the Gershgorin disc is:

$$D_i = \{z \in \mathbb{C} : |z - a_{ii}| \leq R_i\}, \quad R_i = \sum_{j \neq i} |a_{ij}|$$

Interior rows $(1 < i < n)$:

$c_i : a_{ii} = 2$

$r_i : R_i = |-1| + |-1| = 2$

$D_i = \{z : |z-2| \leq 2\}$

Eigenvalues must lie in interval $[2-2, 2+2] = [0,4]$

Boundary rows $(i=1$ and $i=n)$

$c_1 : a_{11} = 2$

$r_1 = |-1| = 1$

$D_1 = D_n = \{z : |z-2| \leq 1\}$

Eigenvalues must lie in interval $[2-1, 2+1] = [1,3]$

$[0,4] \cup [1,3] = [0,4]$. Since all Eigenvalues must lie within this union, all eigenvalues satisfy $\lambda = 0$.

5. a).

$$A^TA = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 9 \end{bmatrix}.$$ Eigenvalues are diagonal entries

$\lambda_1 = 9,\ \lambda_2 = 4,\ \lambda_3 = 1,\ \lambda_4 = 0$; $\sigma_i = \sqrt{\lambda_i}$

$$\boxed{\sigma_1 = 3,\ \sigma_2 = 2,\ \sigma_3 = 1,\ \sigma_4 = 0}$$

b) $A = U\Sigma V^T$. $\Sigma = \begin{bmatrix} 3 & & & 0 \\ & 2 & & \\ & & 1 & \\ 0 & & & 0 \end{bmatrix}$

For $\lambda_1 = 9$, $v_1 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$; For $\lambda = 4$, $v_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}$; For $\lambda = 1$, $v_3 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}$; For $\lambda = 0$, $v_4 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$

So $V = \begin{bmatrix} 1 & & & 0 \\ & 1 & & \\ & & 1 & \\ 0 & & & 1 \end{bmatrix}$

$$Av_1 = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix} v_1 = \begin{bmatrix} 0 \\ 0 \\ 3 \\ 0 \end{bmatrix};\ Av_2 = \begin{bmatrix} 0 \\ 2 \\ 0 \\ 0 \end{bmatrix};\ Av_3 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$u_1 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}$, $u_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}$, $u_3 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$, $u_4 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$ ✓ only remaining one

$$U = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 3 & & & 0 \\ & 2 & & \\ & & 1 & \\ 0 & & & 0 \end{bmatrix} \begin{bmatrix} 0 & & & 1 \\ & & 1 & \\ & 1 & & \\ 1 & & & 0 \end{bmatrix}^T = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix} \checkmark$$

6. a) $A^T A x = A^T b$, $A^T = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 3 & 0 \end{bmatrix} \rightarrow A^T b = 0$

$A^T A = \begin{bmatrix} 0 & & & 0 \\ & 1 & & \\ & & 4 & \\ 0 & & & 9 \end{bmatrix}$

$A^T A x = 0 \rightarrow x_2 = 0,\ x_3 = 0,\ x_4 = 0,\ x_1$ is free $\therefore\ x = \begin{bmatrix} t \\ 0 \\ 0 \\ 0 \end{bmatrix}$

Smallest norm one has $t = 0$ so $\boxed{x^* = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}}$

b) By Eckart-Young theorem the best rank-k approximation of a matrix A is obtained by truncating the SVD sum to first k terms.

For rank $k = 1$, $\hat{A} = \sigma_1 u_1 v_1^T$

The largest singular vector is $\sigma_1 = 3 \Leftrightarrow u_1 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}$

Corresponding right singular vector is $\begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$

$$\tilde{A} = 3 \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

7. By SVD, any $m \times n$ matrix $A = U \Sigma V^T$

$A^T = (U \Sigma V^T)^T = (V^T)^T \Sigma^T U^T = V \Sigma^T U^T$

because $(ABC)^T = C^T B^T A^T$

V is orthogonal

U^T is orthogonal since U is also

Σ^T is transpose of diagonal matrix Σ

Since Σ is diagonal, Σ^T has same diagonal entries (singular values σ :) just with matrix dims flipped.

Therefore the decomposition $A^T = V \Sigma^T U^T$ is a valid SVD for A^T.

Since singular values are the diagonal entries, A and A^T share same singular values

8. a) If B is nonsingular then it's inverse B^{-1} exists.
We can use this to bridge the gap between U and W.

$U = AB \rightarrow A = UB^{-1}$; $W = BC \rightarrow C = B^{-1}W$

$$\boxed{ABC = (UB^{-1})B(B^{-1}W) = UB^{-1}W}$$

b) Since we're given $W = BC$ and B we can solve for some matrix $\tilde{C}$ satisfying $B\tilde{C} = W$. Steps:

- write $W = [w_1 \dots w_s]$ w/ $w_j \in \mathbb{R}^n$
- For each col solve $Bc_j = w_j$ using row reduction, picking any one solution
- Stack solutions: $\tilde{C} = [c_1 \dots c_s]$
- Then we can calculate: $\boxed{ABC = U\tilde{C}}$

Because $U = AB$,

$$U\tilde{C} = (AB)\tilde{C} = A(B\tilde{C}) = AW = A(BC) = ABC$$

Also if $\tilde{C}_1, \tilde{C}_2$ both satisfy $B\tilde{C}_i = W$, then $B(\tilde{C}_1 - \tilde{C}_2) = 0$, so

$$U(\tilde{C}_1 - \tilde{C}_2) = AB(\tilde{C}_1 - \tilde{C}_2) = 0$$

meaning $U\tilde{C}$ is independent of any solution we pick.

We could also instead solve $\tilde{A}B = U$ for any $\tilde{A}$ then find $\tilde{A}W$.